AI Large Model Types and Principles

1. Course Content

- Understand the basic knowledge of AI large models and the types of AI large models used in AI embodied intelligence gameplay.

> [!TIP]
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> - This section is a purely theoretical course. If you need to get started quickly and experience the features, you can skip this section for now.

2. Introduction to AI Large Models

AI large models, i.e., large-scale pre-trained models, are the product of the deep integration of big data, large computing power, and strong algorithms. In simple terms, it is like an intelligent agent that has been "fed" with massive amounts of knowledge and trained repeatedly. By learning from a large amount of data, it masters the laws and patterns in the data, thus possessing powerful versatility and generalization capabilities. This ability allows AI large models to no longer be limited to a single task, but to flexibly apply the knowledge they have learned to solve complex problems in multiple fields, just like humans. It uses multi-source data such as internet text and images to learn data patterns, has strong generalization ability, can adapt to a variety of tasks through transfer learning or prompt engineering, and may also emerge with abilities not set before training, such as logical reasoning and common sense understanding.

- Open Source Model Community: [ModelScope Community]
- Alibaba Bailian Large Model: [Bailian Large Model]

3. Common AI Large Model Categories

3.1 Text Generation Large Models

- Based on the Transformer architecture, they learn the syntax, semantics, and pragmatic rules of language through unsupervised or supervised learning on massive text data, thus being able to generate natural and fluent text based on input prompts or context. Pre-trained through unsupervised learning, they are suitable for tasks such as text generation and dialogue systems; by converting all text tasks into a unified text-to-text problem, they provide a more flexible framework that can handle various tasks such as translation, summarization, and question answering.

- In terms of content creation, they can automatically generate practical texts such as articles, news, and comments, improving content production efficiency, and can also assist authors in creative conception and text polishing.
- In the field of intelligent interaction, they can be applied to intelligent customer service and chatbots to generate natural and fluent replies and enhance user experience.
- In personalized teaching, they can analyze questions, provide explanations of test points, problem-solving ideas, and results, and can also help users with language learning.
- In terms of machine translation, they can achieve automatic translation, and combined with speech models, can also achieve simultaneous interpretation, daily subtitle generation, etc.

3.1.1 Principle Introduction

Based on the Transformer architecture (especially the self-attention mechanism), they model the probability distribution and semantic correlation of language through unsupervised/semi-supervised learning of massive text data to achieve understanding and generation of natural language.

- Pre-training Logic
  - Autoregressive (AR): Such as the GPT series, which predicts the next token through a "causal language model" (e.g., "Today the weather → Today the weather is"), learning the front-to-back dependencies of the text.
  - Autoencoding (AE): Such as BERT, which predicts the masked token through a "masked language model" (e.g., "Today the [mask] is very sunny" → "weather"), learning the bidirectional semantics of the context.
- Key Technologies
  - Attention Mechanism: Dynamically allocates weights to different words in the text to capture long-distance dependencies (e.g., "the event mentioned earlier → the impact later").
  - Prompt Tuning: Activates the specific capabilities of the model through templates (e.g., "Please summarize the following content: {text}") to adapt to downstream tasks.
- Emergent Abilities
  As the parameter scale expands (e.g., to the hundred-billion level), the model may exhibit abilities not explicitly programmed during the pre-training phase, such as logical reasoning, common sense understanding, and few-shot learning.

3.2 Multimodal Large Models

They can process multiple types of input data, such as text, images, audio, video, etc. Through cross-modal learning, they understand the relationships between different modal data and integrate multimodal data to make full use of the information of each modality, building a unified representation space that allows data of different modalities to understand and combine with each other, thereby performing more complex and intelligent tasks. They can be used for cross-modal retrieval, retrieving data of one modality based on data of another modality; in visual question answering, the model answers text questions based on image content; they can also perform image description generation, generating natural language text that describes the content of the image; and can also achieve multimodal dialogue, conducting dialogues involving information of multiple modalities, with broad application prospects in complex environments such as healthcare, transportation, and security monitoring.

3.2.1 Principle Introduction

Through cross-modal alignment and joint modeling, they learn a unified representation space for data of different modalities, achieving semantic correlation and collaborative processing between modalities.

- Contrastive Learning: Such as the CLIP model, which maps the feature vectors of images and text to the same space and trains through "image-text pair matching" (e.g., "picture of a dog → text 'dog'").
- Encoder-Decoder Architecture: Such as DALL-E, where the text encoder extracts semantic features and the image decoder generates the corresponding image (text → image).

Fusion Methods

- Early Fusion: Merges multimodal data at the input layer (e.g., concatenating text embeddings with image pixel features).
- Late Fusion: Processes each modal data separately and fuses the results at the decision layer (e.g., first analyzing text sentiment and image color separately, then making a comprehensive judgment).

3.3 Speech Recognition Models

- Convert input speech signals into text information. Usually based on deep learning algorithms, they first perform feature extraction on the speech signal, and then input the features into a neural network model for training and recognition. The model recognizes different speech patterns and corresponding text content by learning a large amount of speech data. They can assist customer service personnel in quickly recording customer needs and problems, improving service quality and facilitating subsequent inquiries; they can be applied to voice search, freeing up hands, and are suitable for various search environments such as vehicle navigation and mobile phones; they can also convert meeting dialogues into text form, which is convenient for organizing and recording meeting content; in terms of human-computer interaction, they use voice commands to control intelligent facilities, including hardware facilities such as robots and software applications.

  ### 3.3.1 Principle Introduction

  Convert the acoustic features of a speech signal into a text sequence, based on deep learning for end-to-end modeling.

  - Feature Extraction: Pre-process the speech waveform (e.g., framing, windowing) to extract Mel-frequency cepstral coefficients (MFCC) or acoustic feature vectors (e.g., extracted through a CNN).
  - Sequence Modeling: Use a Recurrent Neural Network (RNN/LSTM) or Transformer encoder to capture the temporal dependencies of the speech sequence (e.g., "continuous syllables → words").
  - Decoding and Mapping: Map the feature sequence to a text sequence through Connectionist Temporal Classification (CTC) or an attention mechanism (e.g., acoustic features → "hello").

3.4 Speech Synthesis Models

Convert input text into speech signals. Generally, by training a model to learn the mapping relationship from text to speech, the model generates corresponding speech features based on the input text, and then converts the features into audible speech through speech synthesis technology. Widely used in voice assistants, audiobooks, intelligent customer service and other fields, providing users with voice interaction services, so that devices can communicate with users in a natural and fluent voice.

3.4.1 Principle Introduction

Convert the semantics of text into a natural and fluent speech signal, simulating the rhythm, intonation, and emotion of human pronunciation.

Deep Learning Synthesis:

- Text Analysis: Analyze the semantics, part of speech, and emotion of the text through an NLP model (e.g., "I'm very happy today" → cheerful tone).
- Acoustic Modeling: Use the Tacotron series (encoder-decoder + attention mechanism) to generate the Mel spectrogram of the speech.
- Vocoder: Convert the Mel spectrogram into a waveform signal, such as WaveNet, HiFi-GAN (to improve the naturalness of the speech).

4.AI Model Comparison Summary

Model Type	Input	Output	Core Technology	Typical Scenarios
Natural Language Large Model	Text	Text	Transformer Self-Attention	Writing, Dialogue, Translation
Multimodal Large Model	Text + Image / Audio	Cross-modal Content	Cross-modal Alignment, Joint Encoding	Image-Text Generation, Visual QA
Speech Recognition Model	Speech Waveform	Text	Acoustic Feature Extraction + Sequence Decoding	Meeting Minutes, Voice Search
Speech Synthesis Model	Text	Speech Audio	Text Analysis + Acoustic Modeling + Vocoder	Voice Assistant, Audio Content Production